

Chapitre 7 : Trigonométrie

Professeur : Ismail OUDAHA

Plan de cours

- 1 Cosinus, Sinus et Tangent d'un angle aigu
- 2 Formules trigonométriques
- 3 Relations trigonométriques de deux angles complémentaires

- 1 Cosinus, Sinus et Tangent d'un angle aigu
- 2 Formules trigonométriques
- 3 Relations trigonométriques de deux angles complémentaires

I- Cosinus, Sinus et Tangent d'un angle aigu :

I- Cosinus, Sinus et Tangent d'un angle aigu :

Activité :

I- Cosinus, Sinus et Tangent d'un angle aigu :

Activité :

Soit ABC un triangle rectangle en A tel que :

$$BC = 5 \text{ cm} \quad ; \quad AB = 4 \text{ cm} \quad \text{et} \quad AC = 3 \text{ cm}$$

- ① Calculer les rapports suivants :

$$\frac{AB}{BC} \quad ; \quad \frac{AC}{BC} \quad \text{et} \quad \frac{AC}{AB}$$

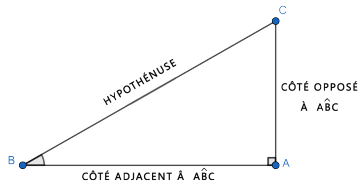
- ② Comparer les rapports obtenus avec 1
- ③ Que remarquez-vous ?

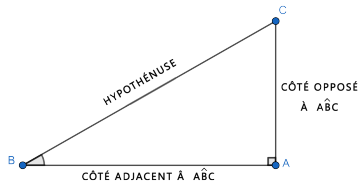
Définition :

Définition :

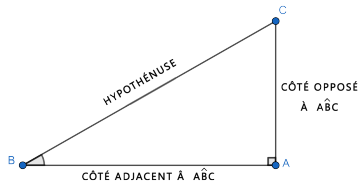
Dans un triangle rectangle :

- **Le cosinus** d'un angle aigu est égale au quotient du côté adjacent sur l'hypoténuse.
- **Le sinus** d'un angle aigu est égale au quotient du côté opposé sur l'hypoténuse.
- **La tangente** d'un angle aigu est égale au quotient du côté opposé sur le côté adjacent.

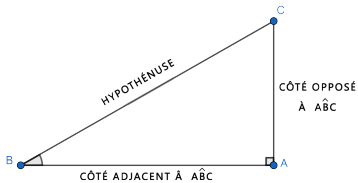




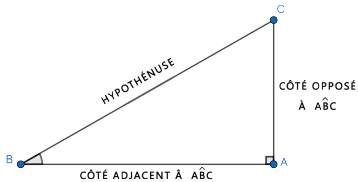
$$\cos \hat{A}BC$$



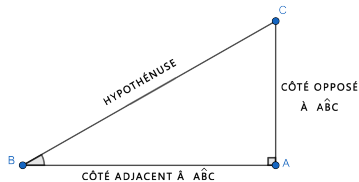
$$\cos \hat{A}BC =$$



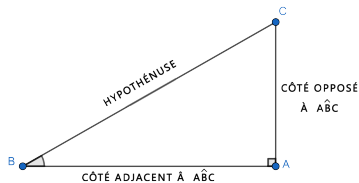
$$\cos \hat{A}BC = \frac{\text{coté adjacent}}{\text{l'hypoténuse}}$$



$$\cos \hat{A}BC = \frac{\text{coté adjacent}}{\text{l'hypoténuse}} =$$

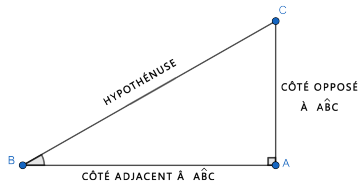


$$\cos \hat{A}BC = \frac{\text{coté adjacent}}{\text{l'hypoténuse}} = \frac{AB}{BC}$$



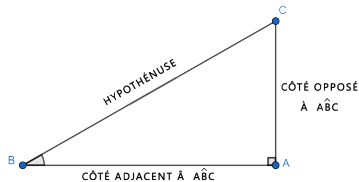
$$\cos \hat{A}BC = \frac{\text{coté adjacent}}{\text{l'hypoténuse}} = \frac{AB}{BC}$$

$$\sin \hat{A}BC$$



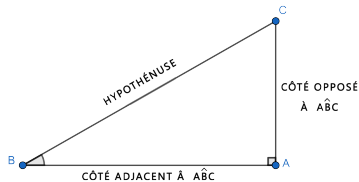
$$\cos \hat{A}BC = \frac{\text{coté adjacent}}{\text{l'hypoténuse}} = \frac{AB}{BC}$$

$$\sin \hat{A}BC =$$



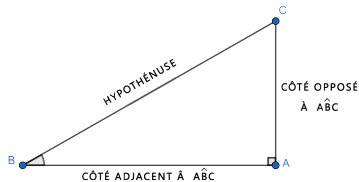
$$\cos \hat{A}BC = \frac{\text{coté adjacent}}{\text{l'hypoténuse}} = \frac{AB}{BC}$$

$$\sin \hat{A}BC = \frac{\text{coté opposé}}{\text{l'hypoténuse}}$$



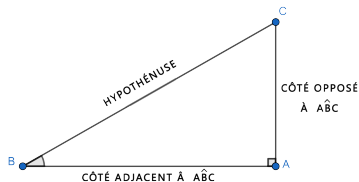
$$\cos \hat{A}BC = \frac{\text{coté adjacent}}{\text{l'hypoténuse}} = \frac{AB}{BC}$$

$$\sin \hat{A}BC = \frac{\text{coté opposé}}{\text{l'hypoténuse}} =$$



$$\cos \hat{A}BC = \frac{\text{coté adjacent}}{\text{l'hypoténuse}} = \frac{AB}{BC}$$

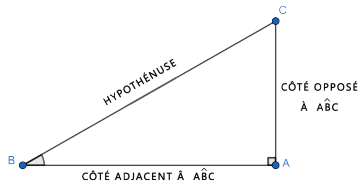
$$\sin \hat{A}BC = \frac{\text{coté opposé}}{\text{l'hypoténuse}} = \frac{AC}{BC}$$



$$\cos \hat{A}BC = \frac{\text{coté adjacent}}{\text{l'hypoténuse}} = \frac{AB}{BC}$$

$$\sin \hat{A}BC = \frac{\text{coté opposé}}{\text{l'hypoténuse}} = \frac{AC}{BC}$$

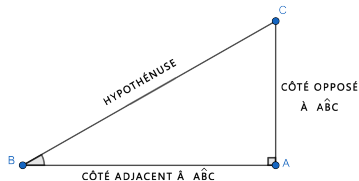
$$\tan \hat{A}BC$$



$$\cos \hat{A}BC = \frac{\text{coté adjacent}}{\text{l'hypoténuse}} = \frac{AB}{BC}$$

$$\sin \hat{A}BC = \frac{\text{coté opposé}}{\text{l'hypoténuse}} = \frac{AC}{BC}$$

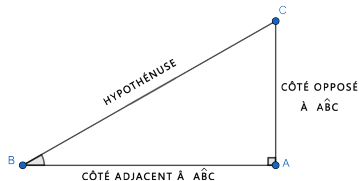
$$\tan \hat{A}BC =$$



$$\cos \hat{A}BC = \frac{\text{coté adjacent}}{\text{l'hypoténuse}} = \frac{AB}{BC}$$

$$\sin \hat{A}BC = \frac{\text{coté opposé}}{\text{l'hypoténuse}} = \frac{AC}{BC}$$

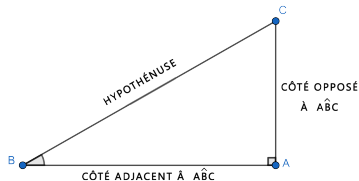
$$\tan \hat{A}BC = \frac{\text{coté opposé}}{\text{coté adjacent}}$$



$$\cos \hat{A}BC = \frac{\text{coté adjacent}}{\text{l'hypoténuse}} = \frac{AB}{BC}$$

$$\sin \hat{A}BC = \frac{\text{coté opposé}}{\text{l'hypoténuse}} = \frac{AC}{BC}$$

$$\tan \hat{A}BC = \frac{\text{coté opposé}}{\text{coté adjacent}} =$$



$$\cos \hat{A}BC = \frac{\text{coté adjacent}}{\text{l'hypoténuse}} = \frac{AB}{BC}$$

$$\sin \hat{A}BC = \frac{\text{coté opposé}}{\text{l'hypoténuse}} = \frac{AC}{BC}$$

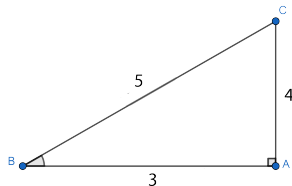
$$\tan \hat{A}BC = \frac{\text{coté opposé}}{\text{coté adjacent}} = \frac{AC}{AB}$$

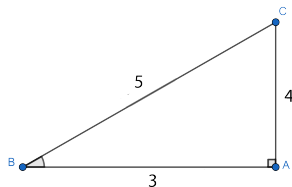
Exemple 1 :

ABC un triangle rectangle en A tel que :

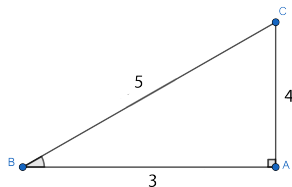
$$AB = 3 \text{ cm} \quad ; \quad AC = 4 \text{ cm} \text{ et } BC = 5 \text{ cm}$$

Calculons : $\cos \hat{A}BC$; $\sin \hat{A}BC$; $\tan \hat{A}BC$

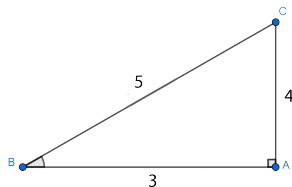




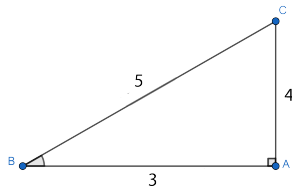
$$\left. \begin{array}{l} \text{Hypotenuse} \\ \text{Adjacent side} \end{array} \right\} \cos \hat{A}BC$$



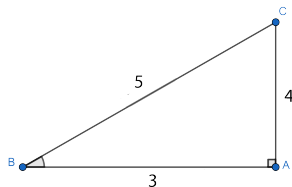
$$\left\{ \cos \hat{A}BC = \right.$$



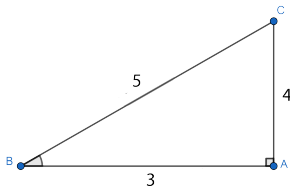
$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} \end{array} \right.$$



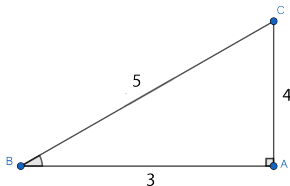
$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} = \end{array} \right.$$



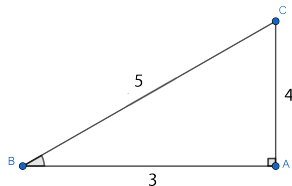
$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} = \frac{3}{5} \end{array} \right.$$



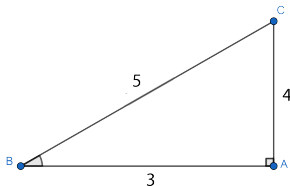
$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} = \frac{3}{5} = \end{array} \right.$$



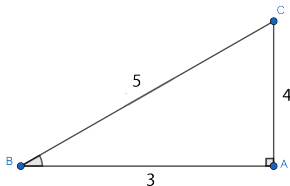
$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} = \frac{3}{5} = 0.6 \end{array} \right.$$



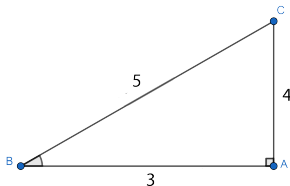
$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} = \frac{3}{5} = 0.6 \\ \sin \hat{ABC} \end{array} \right.$$



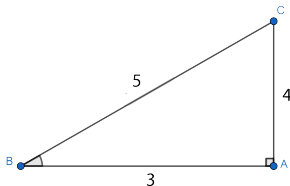
$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} = \frac{3}{5} = 0.6 \\ \sin \hat{ABC} = \end{array} \right.$$



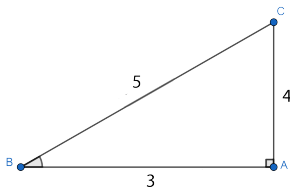
$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} = \frac{3}{5} = 0.6 \\ \sin \hat{ABC} = \frac{AC}{BC} \end{array} \right.$$



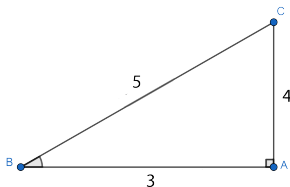
$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} = \frac{3}{5} = 0.6 \\ \sin \hat{ABC} = \frac{AC}{BC} = \end{array} \right.$$



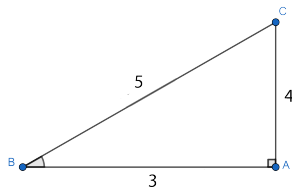
$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} = \frac{3}{5} = 0.6 \\ \sin \hat{ABC} = \frac{AC}{BC} = \frac{4}{5} \end{array} \right.$$



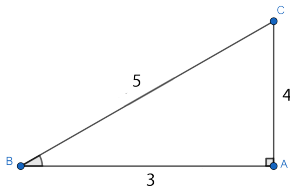
$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} = \frac{3}{5} = 0.6 \\ \sin \hat{ABC} = \frac{AC}{BC} = \frac{4}{5} = \end{array} \right.$$



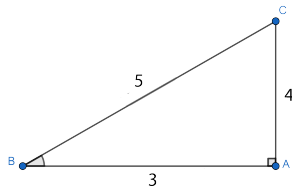
$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} = \frac{3}{5} = 0.6 \\ \sin \hat{ABC} = \frac{AC}{BC} = \frac{4}{5} = 0.8 \end{array} \right.$$



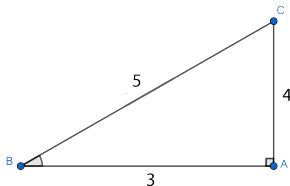
$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} = \frac{3}{5} = 0.6 \\ \sin \hat{ABC} = \frac{AC}{BC} = \frac{4}{5} = 0.8 \\ \tan \hat{ABC} \end{array} \right.$$



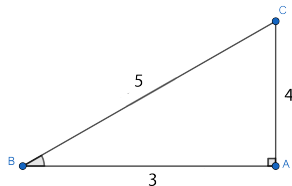
$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} = \frac{3}{5} = 0.6 \\ \sin \hat{ABC} = \frac{AC}{BC} = \frac{4}{5} = 0.8 \\ \tan \hat{ABC} = \end{array} \right.$$



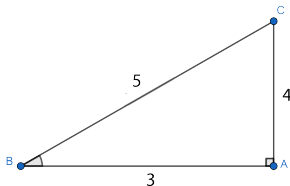
$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} = \frac{3}{5} = 0.6 \\ \sin \hat{ABC} = \frac{AC}{BC} = \frac{4}{5} = 0.8 \\ \tan \hat{ABC} = \frac{AC}{AB} \end{array} \right.$$



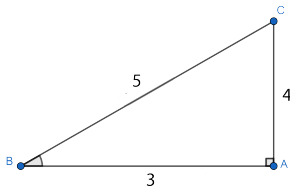
$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} = \frac{3}{5} = 0.6 \\ \sin \hat{ABC} = \frac{AC}{BC} = \frac{4}{5} = 0.8 \\ \tan \hat{ABC} = \frac{AC}{AB} = \end{array} \right.$$



$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} = \frac{3}{5} = 0.6 \\ \sin \hat{ABC} = \frac{AC}{BC} = \frac{4}{5} = 0.8 \\ \tan \hat{ABC} = \frac{AC}{AB} = \frac{4}{3} \end{array} \right.$$



$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} = \frac{3}{5} = 0.6 \\ \sin \hat{ABC} = \frac{AC}{BC} = \frac{4}{5} = 0.8 \\ \tan \hat{ABC} = \frac{AC}{AB} = \frac{4}{3} = \end{array} \right.$$



$$\left\{ \begin{array}{l} \cos \hat{ABC} = \frac{AB}{BC} = \frac{3}{5} = 0.6 \\ \sin \hat{ABC} = \frac{AC}{BC} = \frac{4}{5} = 0.8 \\ \tan \hat{ABC} = \frac{AC}{AB} = \frac{4}{3} = 1.33 \end{array} \right.$$

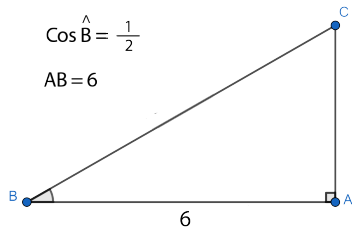
Exemple 2 :

ABC un triangle rectangle en A tel que :

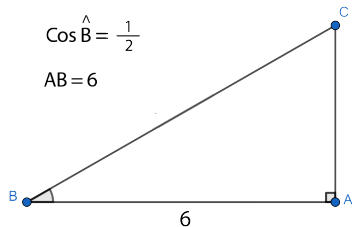
$$AB = 6 \text{ cm} \quad ; \quad \cos \hat{B} = \frac{1}{2}$$

① Calculer : BC , puis déduis AC .

② Calculer : $\sin \hat{B}$; $\tan \hat{B}$

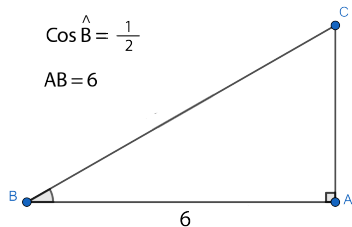


- Calculons BC :



- Calculons BC :

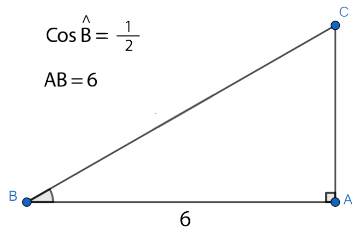
$$\text{On a : } \cos \hat{B} = \frac{AB}{BC}$$

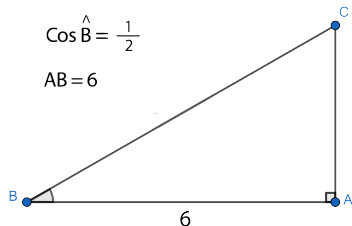


- Calculons BC :

$$\text{On a : } \cos \hat{B} = \frac{AB}{BC}$$

$$\text{càd : } \cos \hat{B} \times BC = AB$$



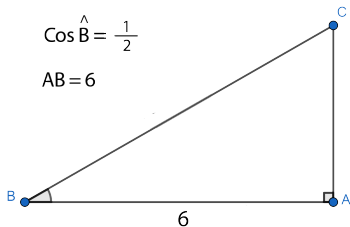


- Calculons BC :

$$\text{On a : } \cos \hat{B} = \frac{AB}{BC}$$

$$\text{càd : } \cos \hat{B} \times BC = AB$$

$$\text{càd : } BC = \frac{AB}{\cos \hat{B}}$$



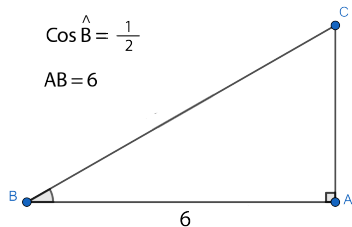
- Calculons BC :

$$\text{On a : } \cos \hat{B} = \frac{AB}{BC}$$

$$\text{càd : } \cos \hat{B} \times BC = AB$$

$$\text{càd : } BC = \frac{AB}{\cos \hat{B}}$$

$$\text{càd : } BC = \frac{6}{\frac{1}{2}}$$



- Calculons BC :

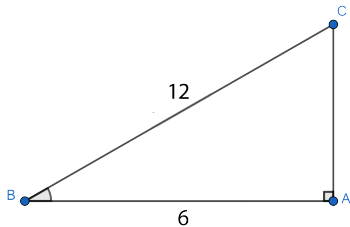
$$\text{On a : } \cos \hat{B} = \frac{AB}{BC}$$

$$\text{càd : } \cos \hat{B} \times BC = AB$$

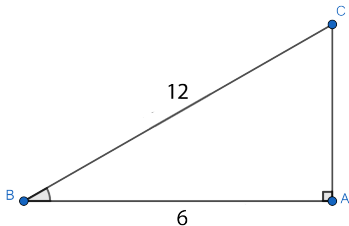
$$\text{càd : } BC = \frac{AB}{\cos \hat{B}}$$

$$\text{càd : } BC = \frac{6}{\frac{1}{2}}$$

$$\text{càd : } BC = 6 \times 2 = 12$$

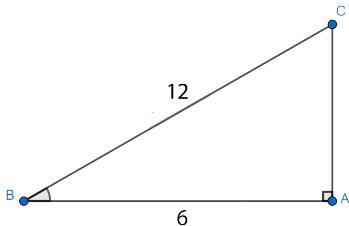


- Calculons AC :



- Calculons AC :

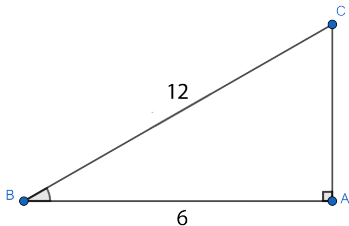
On a le triangle ABC est rectangle en A , donc d'après le théorème de Pythagore , on a :

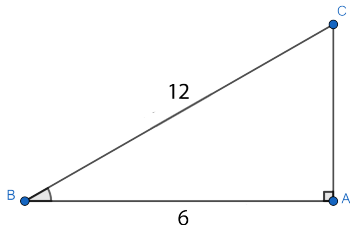


- Calculons AC :

On a le triangle ABC est rectangle en A , donc d'après le théorème de Pythagore , on a :

$$BC^2 = AB^2 + AC^2$$





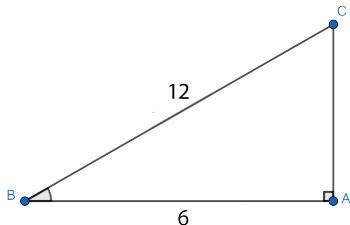
- Calculons AC :

On a le triangle ABC est rectangle en A , donc d'après le théorème de Pythagore , on a :

$$BC^2 = AB^2 + AC^2$$

$$AC^2 = BC^2 - AB^2$$

$$AC^2 = 12^2 - 6^2$$



- Calculons AC :

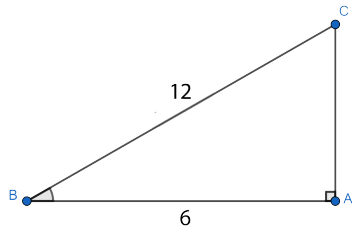
On a le triangle ABC est rectangle en A , donc d'après le théorème de Pythagore , on a :

$$BC^2 = AB^2 + AC^2$$

$$AC^2 = BC^2 - AB^2$$

$$AC^2 = 12^2 - 6^2$$

$$AC^2 = 144 - 36$$



- Calculons AC :

On a le triangle ABC est rectangle en A , donc d'après le théorème de Pythagore , on a :

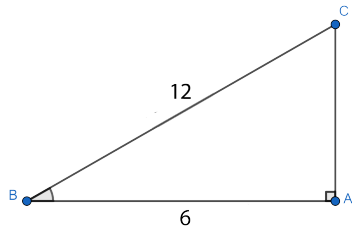
$$BC^2 = AB^2 + AC^2$$

$$AC^2 = BC^2 - AB^2$$

$$AC^2 = 12^2 - 6^2$$

$$AC^2 = 144 - 36$$

$$AC^2 = 108$$



- Calculons AC :

On a le triangle ABC est rectangle en A , donc d'après le théorème de Pythagore , on a :

$$BC^2 = AB^2 + AC^2$$

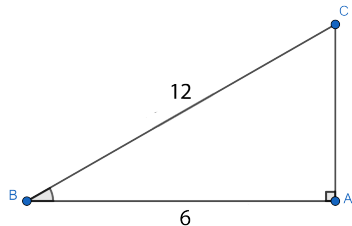
$$AC^2 = BC^2 - AB^2$$

$$AC^2 = 12^2 - 6^2$$

$$AC^2 = 144 - 36$$

$$AC^2 = 108$$

$$AC = \sqrt{108}$$



- Calculons AC :

On a le triangle ABC est rectangle en A , donc d'après le théorème de Pythagore , on a :

$$BC^2 = AB^2 + AC^2$$

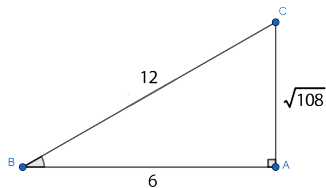
$$AC^2 = BC^2 - AB^2$$

$$AC^2 = 12^2 - 6^2$$

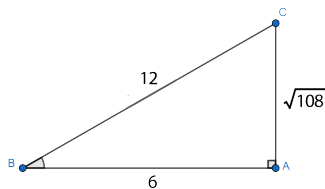
$$AC^2 = 144 - 36$$

$$AC^2 = 108$$

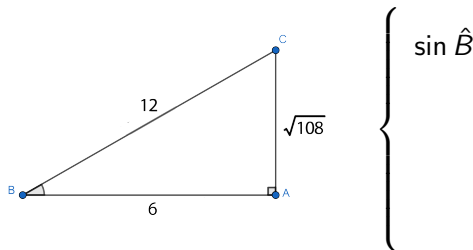
$$AC = \sqrt{108}$$



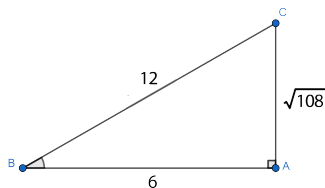
- Calculons $\sin \hat{B}$ et $\tan \hat{B}$:



- Calculons $\sin \hat{B}$ et $\tan \hat{B}$:

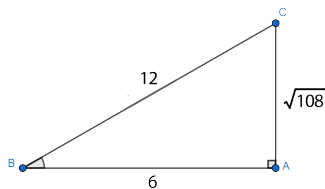


- Calculons $\sin \hat{B}$ et $\tan \hat{B}$:



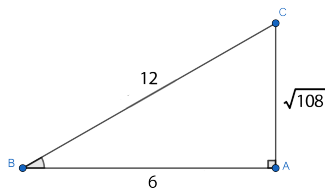
$$\left\{ \begin{array}{l} \sin \hat{B} = \\ \tan \hat{B} = \end{array} \right.$$

- Calculons $\sin \hat{B}$ et $\tan \hat{B}$:



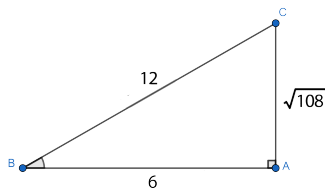
$$\left\{ \begin{array}{l} \sin \hat{B} = \frac{AC}{BC} \end{array} \right.$$

- Calculons $\sin \hat{B}$ et $\tan \hat{B}$:



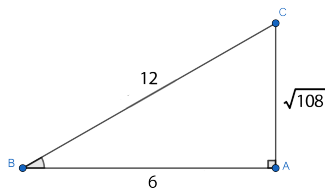
$$\left\{ \begin{array}{l} \sin \hat{B} = \frac{AC}{BC} = \end{array} \right.$$

- Calculons $\sin \hat{B}$ et $\tan \hat{B}$:



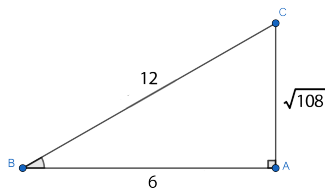
$$\left\{ \begin{array}{l} \sin \hat{B} = \frac{AC}{BC} = \frac{\sqrt{108}}{12} \end{array} \right.$$

- Calculons $\sin \hat{B}$ et $\tan \hat{B}$:



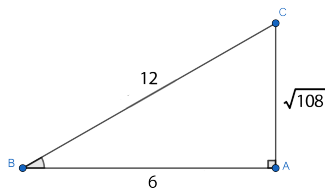
$$\left\{ \begin{array}{l} \sin \hat{B} = \frac{AC}{BC} = \frac{\sqrt{108}}{12} = \end{array} \right.$$

- Calculons $\sin \hat{B}$ et $\tan \hat{B}$:



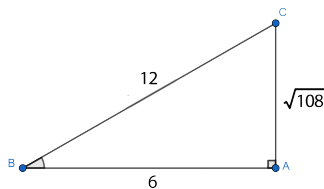
$$\left\{ \begin{array}{l} \sin \hat{B} = \frac{AC}{BC} = \frac{\sqrt{108}}{12} = 0.86 \end{array} \right.$$

- Calculons $\sin \hat{B}$ et $\tan \hat{B}$:



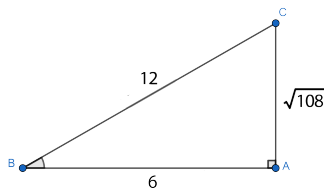
$$\left\{ \begin{array}{l} \sin \hat{B} = \frac{AC}{BC} = \frac{\sqrt{108}}{12} = 0.86 \\ \tan \hat{B} \end{array} \right.$$

- Calculons $\sin \hat{B}$ et $\tan \hat{B}$:



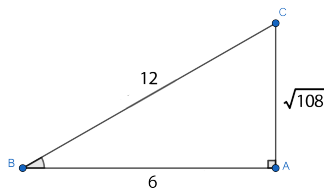
$$\left\{ \begin{array}{l} \sin \hat{B} = \frac{AC}{BC} = \frac{\sqrt{108}}{12} = 0.86 \\ \tan \hat{B} = \end{array} \right.$$

- Calculons $\sin \hat{B}$ et $\tan \hat{B}$:



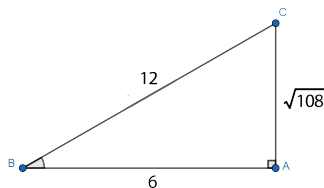
$$\left\{ \begin{array}{l} \sin \hat{B} = \frac{AC}{BC} = \frac{\sqrt{108}}{12} = 0.86 \\ \tan \hat{B} = \frac{AC}{AB} \end{array} \right.$$

- Calculons $\sin \hat{B}$ et $\tan \hat{B}$:



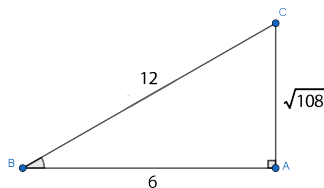
$$\left\{ \begin{array}{l} \sin \hat{B} = \frac{AC}{BC} = \frac{\sqrt{108}}{12} = 0.86 \\ \tan \hat{B} = \frac{AC}{AB} = \end{array} \right.$$

- Calculons $\sin \hat{B}$ et $\tan \hat{B}$:



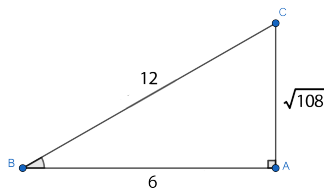
$$\left\{ \begin{array}{l} \sin \hat{B} = \frac{AC}{BC} = \frac{\sqrt{108}}{12} = 0.86 \\ \tan \hat{B} = \frac{AC}{AB} = \frac{\sqrt{108}}{6} \end{array} \right.$$

- Calculons $\sin \hat{B}$ et $\tan \hat{B}$:



$$\left\{ \begin{array}{l} \sin \hat{B} = \frac{AC}{BC} = \frac{\sqrt{108}}{12} = 0.86 \\ \tan \hat{B} = \frac{AC}{AB} = \frac{\sqrt{108}}{6} = \end{array} \right.$$

- Calculons $\sin \hat{B}$ et $\tan \hat{B}$:



$$\left\{ \begin{array}{l} \sin \hat{B} = \frac{AC}{BC} = \frac{\sqrt{108}}{12} = 0.86 \\ \tan \hat{B} = \frac{AC}{AB} = \frac{\sqrt{108}}{6} = 1.73 \end{array} \right.$$

Remarque :

- Le cosinus et le sinus d'un angle aigu α sont toujours compris entre 0 et 1 et n'ont pas d'unité.

Remarque :

- Le cosinus et le sinus d'un angle aigu α sont toujours compris entre 0 et 1 et n'ont pas d'unité.

$$0 \leq \cos \alpha \leq 1 \quad ; \quad 0 \leq \sin \alpha \leq 1$$

Remarque :

- Le cosinus et le sinus d'un angle aigu α sont toujours compris entre 0 et 1 et n'ont pas d'unité.

$$0 \leq \cos \alpha \leq 1 \quad ; \quad 0 \leq \sin \alpha \leq 1$$

- La tangente d'un angle aigu α est un nombre strictement positif et n'a pas d'unité.

Remarque :

- Le cosinus et le sinus d'un angle aigu α sont toujours compris entre 0 et 1 et n'ont pas d'unité.

$$0 \leq \cos \alpha \leq 1 \quad ; \quad 0 \leq \sin \alpha \leq 1$$

- La tangente d'un angle aigu α est un nombre strictement positif et n'a pas d'unité.

$$\tan \alpha > 0$$

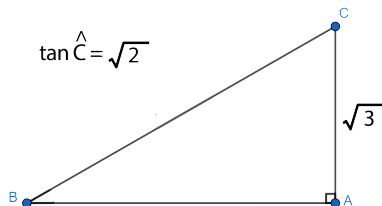
Application :

ABC un triangle rectangle en A tel que :

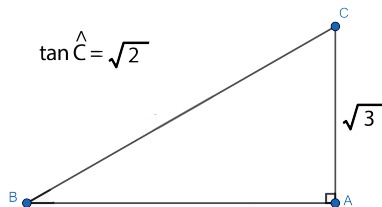
$$AC = \sqrt{3} \text{ cm} \quad ; \quad \tan \hat{C} = \sqrt{2}$$

❶ Calculer : AB , puis déduis BC .

❷ Calculer : $\sin \hat{C}$; $\tan \hat{C}$

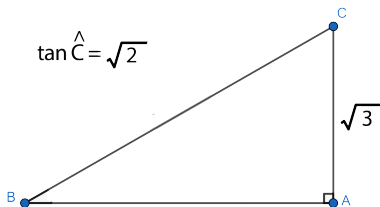


- Calculons AB :



- Calculons AB :

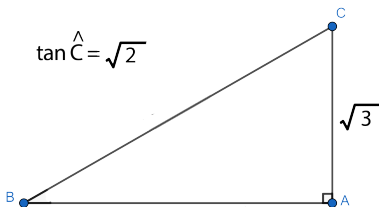
$$\text{On a : } \tan \hat{C} = \frac{AB}{AC}$$



- Calculons AB :

$$\text{On a : } \tan \hat{C} = \frac{AB}{AC}$$

$$\text{càd : } AB = \tan \hat{C} \times AC$$

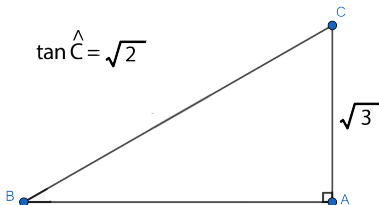


- Calculons AB :

$$\text{On a : } \tan \hat{C} = \frac{AB}{AC}$$

$$\text{càd : } AB = \tan \hat{C} \times AC$$

$$\text{càd : } AB = \sqrt{2} \times \sqrt{3}$$



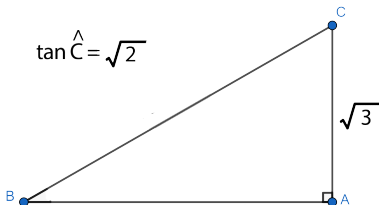
- Calculons AB :

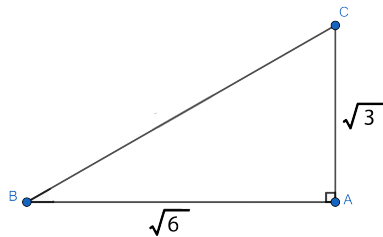
$$\text{On a : } \tan \hat{C} = \frac{AB}{AC}$$

$$\text{càd : } AB = \tan \hat{C} \times AC$$

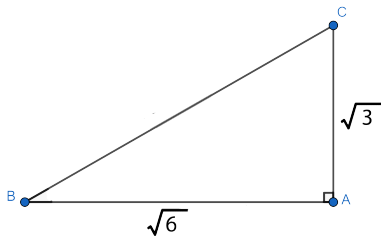
$$\text{càd : } AB = \sqrt{2} \times \sqrt{3}$$

$$\text{càd : } AB = \sqrt{6}$$



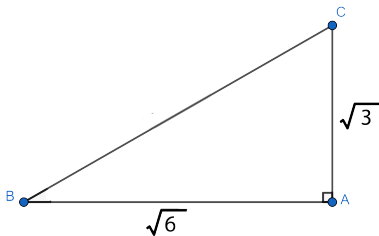


- Calculons BC :



- Calculons BC :

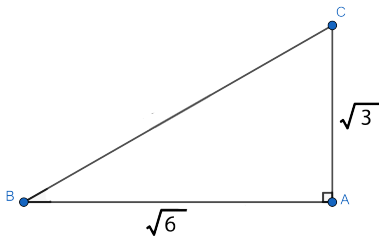
On a le triangle ABC est rectangle en A , donc d'après le théorème de Pythagore , on a :



- Calculons BC :

On a le triangle ABC est rectangle en A , donc d'après le théorème de Pythagore , on a :

$$BC^2 = AB^2 + AC^2$$

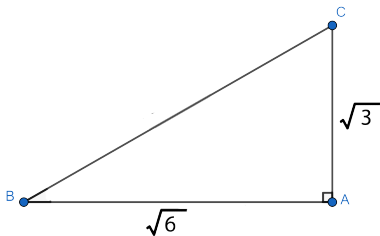


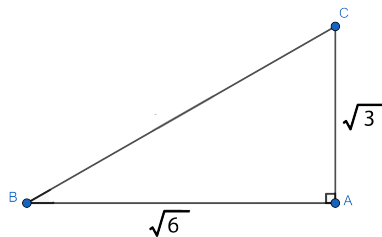
- Calculons BC :

On a le triangle ABC est rectangle en A , donc d'après le théorème de Pythagore , on a :

$$BC^2 = AB^2 + AC^2$$

$$BC^2 = \sqrt{6}^2 + \sqrt{3}^2$$





- Calculons BC :

On a le triangle ABC est rectangle en A , donc d'après le théorème de Pythagore , on a :

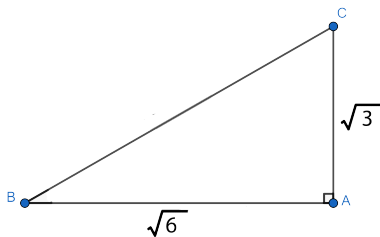
$$BC^2 = AB^2 + AC^2$$

$$BC^2 = \sqrt{6}^2 + \sqrt{3}^2$$

$$BC^2 = 6 + 3$$

- Calculons BC :

On a le triangle ABC est rectangle en A , donc d'après le théorème de Pythagore , on a :



$$BC^2 = AB^2 + AC^2$$

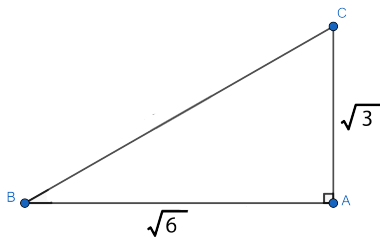
$$BC^2 = \sqrt{6}^2 + \sqrt{3}^2$$

$$BC^2 = 6 + 3$$

$$BC^2 = 9$$

- Calculons BC :

On a le triangle ABC est rectangle en A , donc d'après le théorème de Pythagore , on a :



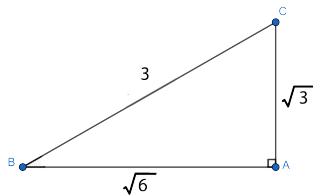
$$BC^2 = AB^2 + AC^2$$

$$BC^2 = \sqrt{6}^2 + \sqrt{3}^2$$

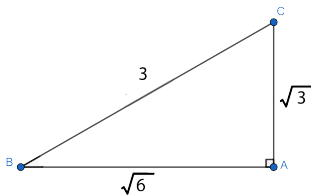
$$BC^2 = 6 + 3$$

$$BC^2 = 9$$

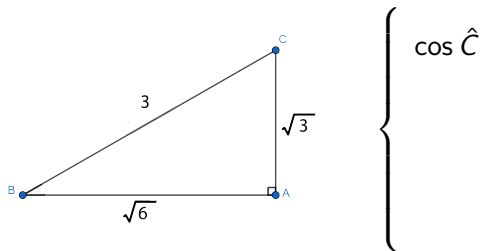
$$BC = \sqrt{9} = 3$$



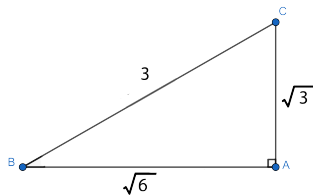
- Calculons $\cos \hat{C}$ et $\sin \hat{C}$:



- Calculons $\cos \hat{C}$ et $\sin \hat{C}$:

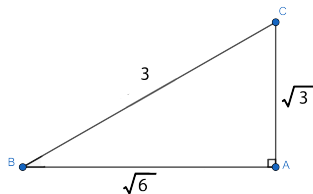


- Calculons $\cos \hat{C}$ et $\sin \hat{C}$:



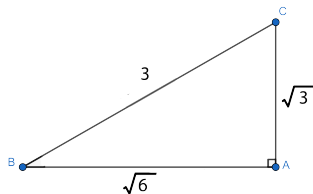
$$\left\{ \begin{array}{l} \cos \hat{C} = \end{array} \right.$$

- Calculons $\cos \hat{C}$ et $\sin \hat{C}$:



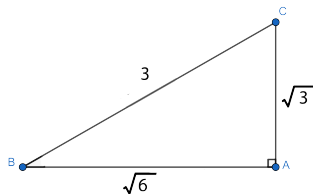
$$\left\{ \begin{array}{l} \cos \hat{C} = \frac{AC}{BC} \end{array} \right.$$

- Calculons $\cos \hat{C}$ et $\sin \hat{C}$:



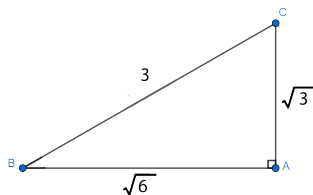
$$\left\{ \begin{array}{l} \cos \hat{C} = \frac{AC}{BC} = \end{array} \right.$$

- Calculons $\cos \hat{C}$ et $\sin \hat{C}$:



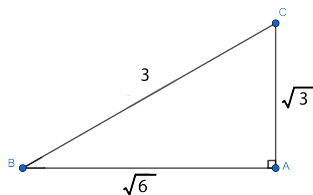
$$\left\{ \begin{array}{l} \cos \hat{C} = \frac{AC}{BC} = \frac{\sqrt{3}}{3} \end{array} \right.$$

- Calculons $\cos \hat{C}$ et $\sin \hat{C}$:



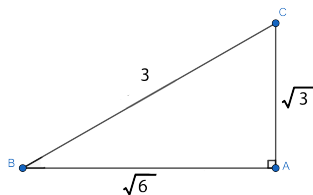
$$\left\{ \begin{array}{l} \cos \hat{C} = \frac{AC}{BC} = \frac{\sqrt{3}}{3} = \end{array} \right.$$

- Calculons $\cos \hat{C}$ et $\sin \hat{C}$:



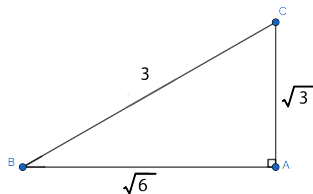
$$\left\{ \begin{array}{l} \cos \hat{C} = \frac{AC}{BC} = \frac{\sqrt{3}}{3} = 0.57 \end{array} \right.$$

- Calculons $\cos \hat{C}$ et $\sin \hat{C}$:



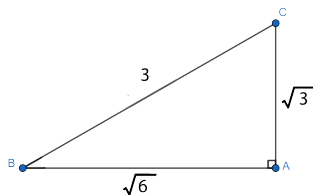
$$\left\{ \begin{array}{l} \cos \hat{C} = \frac{AC}{BC} = \frac{\sqrt{3}}{3} = 0.57 \\ \sin \hat{C} \end{array} \right.$$

- Calculons $\cos \hat{C}$ et $\sin \hat{C}$:



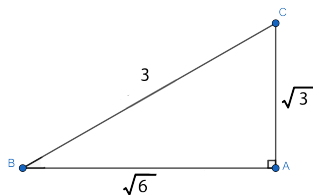
$$\left\{ \begin{array}{l} \cos \hat{C} = \frac{AC}{BC} = \frac{\sqrt{3}}{3} = 0.57 \\ \sin \hat{C} = \end{array} \right.$$

- Calculons $\cos \hat{C}$ et $\sin \hat{C}$:



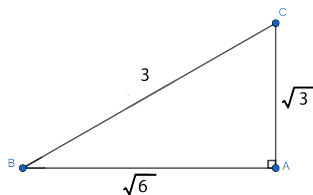
$$\left\{ \begin{array}{l} \cos \hat{C} = \frac{AC}{BC} = \frac{\sqrt{3}}{3} = 0.57 \\ \sin \hat{C} = \frac{AB}{BC} \end{array} \right.$$

- Calculons $\cos \hat{C}$ et $\sin \hat{C}$:



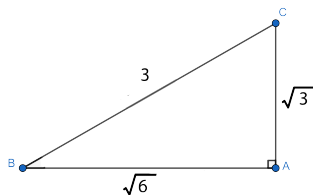
$$\left\{ \begin{array}{l} \cos \hat{C} = \frac{AC}{BC} = \frac{\sqrt{3}}{3} = 0.57 \\ \sin \hat{C} = \frac{AB}{BC} = \end{array} \right.$$

- Calculons $\cos \hat{C}$ et $\sin \hat{C}$:



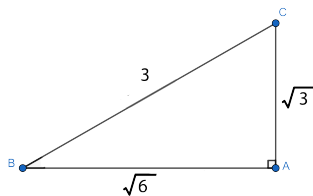
$$\left\{ \begin{array}{l} \cos \hat{C} = \frac{AC}{BC} = \frac{\sqrt{3}}{3} = 0.57 \\ \sin \hat{C} = \frac{AB}{BC} = \frac{\sqrt{6}}{3} \end{array} \right.$$

- Calculons $\cos \hat{C}$ et $\sin \hat{C}$:



$$\left\{ \begin{array}{l} \cos \hat{C} = \frac{AC}{BC} = \frac{\sqrt{3}}{3} = 0.57 \\ \sin \hat{C} = \frac{AB}{BC} = \frac{\sqrt{6}}{3} = \end{array} \right.$$

- Calculons $\cos \hat{C}$ et $\sin \hat{C}$:



$$\left\{ \begin{array}{l} \cos \hat{C} = \frac{AC}{BC} = \frac{\sqrt{3}}{3} = 0.57 \\ \sin \hat{C} = \frac{AB}{BC} = \frac{\sqrt{6}}{3} = 0.81 \end{array} \right.$$

Application 1 :

ABC un triangle rectangle en A tel que :

$$AC = \sqrt{3} \text{ cm} \quad ; \quad \tan \hat{C} = \sqrt{2}$$

❶ Calculer : AB , puis déduis BC .

❷ Calculer : $\sin \hat{C}$; $\tan \hat{C}$

Application 2 :

ABC un triangle rectangle en A tel que :

$$AB = 8 \text{ cm} \quad ; \quad \cos \hat{B} = \frac{\sqrt{3}}{2}$$

❶ Calculer : BC , puis déduis AC .

❷ Calculer : $\sin \hat{B}$; $\tan \hat{B}$

- 1 Cosinus, Sinus et Tangent d'un angle aigu
- 2 **Formules trigonométriques**
- 3 Relations trigonométriques de deux angles complémentaires

II- Formules trigonométriques :

II- Formules trigonométriques :

Activité :

II- Formules trigonométriques :

Activité :

Soit ABC un triangle rectangle en A .

❶ Montrer que : $\cos^2(\widehat{ABC}) + \sin^2(\widehat{ABC}) = 1$

❷ Montrer que : $\tan(\widehat{ABC}) = \frac{\sin(\widehat{ABC})}{\cos(\widehat{ABC})}$

Propriété :

Propriété :

Pour tout angle aigu α on a :

$$\cos^2 \alpha + \sin^2 \alpha = 1 \quad ; \quad \tan \alpha = \frac{\sin \alpha}{\cos \alpha}$$

Propriété :

Pour tout angle aigu α on a :

$$\cos^2 \alpha + \sin^2 \alpha = 1 \quad ; \quad \tan \alpha = \frac{\sin \alpha}{\cos \alpha}$$

Remarque :

Propriété :

Pour tout angle aigu α on a :

$$\cos^2 \alpha + \sin^2 \alpha = 1 \quad ; \quad \tan \alpha = \frac{\sin \alpha}{\cos \alpha}$$

Remarque :

- La première formule permet de calculer $\sin \alpha$ connaissant $\cos \alpha$ (ou de calculer $\cos \alpha$ connaissant $\sin \alpha$).

Propriété :

Pour tout angle aigu α on a :

$$\cos^2 \alpha + \sin^2 \alpha = 1 \quad ; \quad \tan \alpha = \frac{\sin \alpha}{\cos \alpha}$$

Remarque :

- La première formule permet de calculer $\sin \alpha$ connaissant $\cos \alpha$ (ou de calculer $\cos \alpha$ connaissant $\sin \alpha$).
- La deuxième formule permet de calculer $\cos \alpha$, $\sin \alpha$ ou $\tan \alpha$ connaissant deux de ces nombres.

Exemple 1 :

Soit α un angle aigu tel que : $\cos \alpha = 0,8$.

calculer : $\sin \alpha$ et $\tan \alpha$.

Exemple 1 :

Soit α un angle aigu tel que : $\cos \alpha = 0,8$.

calculer : $\sin \alpha$ et $\tan \alpha$.

- Calculons $\sin \alpha$:

Exemple 1 :

Soit α un angle aigu tel que : $\cos \alpha = 0,8$.
calculer : $\sin \alpha$ et $\tan \alpha$.

- Calculons $\sin \alpha$:

On sait que : $\cos^2 \alpha + \sin^2 \alpha = 1$

Exemple 1 :

Soit α un angle aigu tel que : $\cos \alpha = 0,8$.
calculer : $\sin \alpha$ et $\tan \alpha$.

- Calculons $\sin \alpha$:

On sait que : $\cos^2 \alpha + \sin^2 \alpha = 1$

Alors : $\sin^2 \alpha = 1 - \cos^2 \alpha$

Exemple 1 :

Soit α un angle aigu tel que : $\cos \alpha = 0,8$.
calculer : $\sin \alpha$ et $\tan \alpha$.

- Calculons $\sin \alpha$:

On sait que : $\cos^2 \alpha + \sin^2 \alpha = 1$

Alors : $\sin^2 \alpha = 1 - \cos^2 \alpha$

c'est à dire : $\sin^2 \alpha = 1 - 0,8^2$

Exemple 1 :

Soit α un angle aigu tel que : $\cos \alpha = 0,8$.
calculer : $\sin \alpha$ et $\tan \alpha$.

- Calculons $\sin \alpha$:

On sait que : $\cos^2 \alpha + \sin^2 \alpha = 1$

Alors : $\sin^2 \alpha = 1 - \cos^2 \alpha$

c'est à dire : $\sin^2 \alpha = 1 - 0,8^2$

c'est à dire : $\sin^2 \alpha = 1 - 0,64$

Exemple 1 :

Soit α un angle aigu tel que : $\cos \alpha = 0,8$.
calculer : $\sin \alpha$ et $\tan \alpha$.

- Calculons $\sin \alpha$:

On sait que : $\cos^2 \alpha + \sin^2 \alpha = 1$

Alors : $\sin^2 \alpha = 1 - \cos^2 \alpha$

c'est à dire : $\sin^2 \alpha = 1 - 0,8^2$

c'est à dire : $\sin^2 \alpha = 1 - 0,64$

c'est à dire : $\sin^2 \alpha = 0,36$

Exemple 1 :

Soit α un angle aigu tel que : $\cos \alpha = 0,8$.
calculer : $\sin \alpha$ et $\tan \alpha$.

- Calculons $\sin \alpha$:

On sait que : $\cos^2 \alpha + \sin^2 \alpha = 1$

Alors : $\sin^2 \alpha = 1 - \cos^2 \alpha$

c'est à dire : $\sin^2 \alpha = 1 - 0,8^2$

c'est à dire : $\sin^2 \alpha = 1 - 0,64$

c'est à dire : $\sin^2 \alpha = 0,36$

c'est à dire : $\sin \alpha = \sqrt{0,36}$

Exemple 1 :

Soit α un angle aigu tel que : $\cos \alpha = 0,8$.
calculer : $\sin \alpha$ et $\tan \alpha$.

- Calculons $\sin \alpha$:

On sait que : $\cos^2 \alpha + \sin^2 \alpha = 1$

Alors : $\sin^2 \alpha = 1 - \cos^2 \alpha$

c'est à dire : $\sin^2 \alpha = 1 - 0,8^2$

c'est à dire : $\sin^2 \alpha = 1 - 0,64$

c'est à dire : $\sin^2 \alpha = 0,36$

c'est à dire : $\sin \alpha = \sqrt{0,36}$

Donc : $\sin \alpha = 0,6$

- Calculons $\tan \alpha$:

- Calculons $\tan \alpha$:

On sait que : $\tan \alpha = \frac{\sin \alpha}{\cos \alpha}$

- Calculons $\tan \alpha$:

On sait que : $\tan \alpha = \frac{\sin \alpha}{\cos \alpha}$

C'est à dire : $\tan \alpha = \frac{0,6}{0,8}$

- Calculons $\tan \alpha$:

On sait que : $\tan \alpha = \frac{\sin \alpha}{\cos \alpha}$

C'est à dire : $\tan \alpha = \frac{0,6}{0,8}$

Donc : $\tan \alpha = 0,75$

Exemple 2 :

Exemple 2 :

$$M = 2 \cos^2 \alpha + 3 \sin^2 \alpha - 2$$

Exemple 2 :

$$M = 2 \cos^2 \alpha + 3 \sin^2 \alpha - 2$$

$$= 2 \cos^2 \alpha + 2 \sin^2 \alpha + \sin^2 \alpha - 2$$

Exemple 2 :

$$M = 2 \cos^2 \alpha + 3 \sin^2 \alpha - 2$$

$$= 2 \cos^2 \alpha + 2 \sin^2 \alpha + \sin^2 \alpha - 2$$

$$= 2 (\cos^2 \alpha + \sin^2 \alpha) + \sin^2 \alpha - 2$$

Exemple 2 :

$$M = 2 \cos^2 \alpha + 3 \sin^2 \alpha - 2$$

$$= 2 \cos^2 \alpha + 2 \sin^2 \alpha + \sin^2 \alpha - 2$$

$$= 2 (\cos^2 \alpha + \sin^2 \alpha) + \sin^2 \alpha - 2$$

$$= 2 \times 1 + \sin^2 \alpha - 2$$

Exemple 2 :

$$\begin{aligned}M &= 2 \cos^2 \alpha + 3 \sin^2 \alpha - 2 \\&= 2 \cos^2 \alpha + 2 \sin^2 \alpha + \sin^2 \alpha - 2 \\&= 2 (\cos^2 \alpha + \sin^2 \alpha) + \sin^2 \alpha - 2 \\&= 2 \times 1 + \sin^2 \alpha - 2 \\&= 2 + \sin^2 \alpha - 2\end{aligned}$$

Exemple 2 :

$$\begin{aligned}M &= 2 \cos^2 \alpha + 3 \sin^2 \alpha - 2 \\&= 2 \cos^2 \alpha + 2 \sin^2 \alpha + \sin^2 \alpha - 2 \\&= 2 (\cos^2 \alpha + \sin^2 \alpha) + \sin^2 \alpha - 2 \\&= 2 \times 1 + \sin^2 \alpha - 2 \\&= 2 + \sin^2 \alpha - 2 \\&= \sin^2 \alpha\end{aligned}$$

Exemple 3 :

Exemple 3 :

$$N = \sqrt{1 + \cos \alpha} \times \sqrt{1 - \cos \alpha} \times \frac{1}{\sin \alpha}$$

Exemple 3 :

$$\begin{aligned} N &= \sqrt{1 + \cos \alpha} \times \sqrt{1 - \cos \alpha} \times \frac{1}{\sin \alpha} \\ &= \sqrt{(1 + \cos \alpha) \times (1 - \cos \alpha)} \times \frac{1}{\sin \alpha} \end{aligned}$$

Exemple 3 :

$$\begin{aligned} N &= \sqrt{1 + \cos \alpha} \times \sqrt{1 - \cos \alpha} \times \frac{1}{\sin \alpha} \\ &= \sqrt{(1 + \cos \alpha) \times (1 - \cos \alpha)} \times \frac{1}{\sin \alpha} \\ &= \sqrt{1 - \cos^2 \alpha} \times \frac{1}{\sin \alpha} \end{aligned}$$

Exemple 3 :

$$\begin{aligned} N &= \sqrt{1 + \cos \alpha} \times \sqrt{1 - \cos \alpha} \times \frac{1}{\sin \alpha} \\ &= \sqrt{(1 + \cos \alpha) \times (1 - \cos \alpha)} \times \frac{1}{\sin \alpha} \\ &= \sqrt{1 - \cos^2 \alpha} \times \frac{1}{\sin \alpha} \\ &= \sqrt{\sin^2 \alpha} \times \frac{1}{\sin \alpha} \end{aligned}$$

Exemple 3 :

$$\begin{aligned} N &= \sqrt{1 + \cos \alpha} \times \sqrt{1 - \cos \alpha} \times \frac{1}{\sin \alpha} \\ &= \sqrt{(1 + \cos \alpha) \times (1 - \cos \alpha)} \times \frac{1}{\sin \alpha} \\ &= \sqrt{1 - \cos^2 \alpha} \times \frac{1}{\sin \alpha} \\ &= \sqrt{\sin^2 \alpha} \times \frac{1}{\sin \alpha} \\ &= \sin \alpha \times \frac{1}{\sin \alpha} \end{aligned}$$

Exemple 3 :

$$\begin{aligned} N &= \sqrt{1 + \cos \alpha} \times \sqrt{1 - \cos \alpha} \times \frac{1}{\sin \alpha} \\ &= \sqrt{(1 + \cos \alpha) \times (1 - \cos \alpha)} \times \frac{1}{\sin \alpha} \\ &= \sqrt{1 - \cos^2 \alpha} \times \frac{1}{\sin \alpha} \\ &= \sqrt{\sin^2 \alpha} \times \frac{1}{\sin \alpha} \\ &= \sin \alpha \times \frac{1}{\sin \alpha} \\ &= 1 \end{aligned}$$

Application :

Application :

① Sachant que : $\cos \alpha = \frac{3}{5}$. Calculer $\sin \alpha$ et $\tan \alpha$

② Sachant que : $\sin \alpha = \frac{\sqrt{3}}{2}$. Calculer $\cos \alpha$ et $\tan \alpha$

- ① Cosinus, Sinus et Tangent d'un angle aigu
- ② Formules trigonométriques
- ③ Relations trigonométriques de deux angles complémentaires

III- Relations trigonométriques de deux angles complémentaires :

III- Relations trigonométriques de deux angles complémentaires :

Activité :

III- Relations trigonométriques de deux angles complémentaires :

Activité :

Soit ABC un triangle rectangle en A .

- 1 Calculer $\widehat{ACB} + \widehat{ABC}$
- 2 Déterminer les rapports trigonométriques de $\widehat{ABC}$.
- 3 Déterminer les rapports trigonométriques de $\widehat{ACB}$.
- 4 Que remarquez-vous ?

Propriété :

Propriété :

Si α et β sont les mesures de deux angles complémentaires, alors :

$$\cos \alpha = \sin \beta \quad ; \quad \sin \alpha = \cos \beta \quad ; \quad \tan \alpha = \frac{1}{\tan \beta}$$

Exemple 1 :

Exemple 1 :

Soient $\alpha = 30^\circ$ et $\beta = 60^\circ$ deux angles complémentaires, alors :

Exemple 1 :

Soient $\alpha = 30^\circ$ et $\beta = 60^\circ$ deux angles complémentaires, alors :

$$\cos(30^\circ) = \sin(60^\circ)$$

Exemple 1 :

Soient $\alpha = 30^\circ$ et $\beta = 60^\circ$ deux angles complémentaires, alors :

$$\cos(30^\circ) = \sin(60^\circ)$$

$$\sin(30^\circ) = \cos(60^\circ)$$

Exemple 1 :

Soient $\alpha = 30^\circ$ et $\beta = 60^\circ$ deux angles complémentaires, alors :

$$\cos(30^\circ) = \sin(60^\circ)$$

$$\sin(30^\circ) = \cos(60^\circ)$$

$$\tan(30^\circ) = \frac{1}{\tan(60^\circ)}$$

Exemple 2 :

Exemple 2 :

$$A = \cos^2(10^\circ) + \sin^2(40^\circ) + \cos^2(80^\circ) + \sin^2(50^\circ)$$

Exemple 2 :

$$\begin{aligned} A &= \cos^2(10^\circ) + \sin^2(40^\circ) + \cos^2(80^\circ) + \sin^2(50^\circ) \\ &= \sin^2(80^\circ) + \cos^2(50^\circ) + \cos^2(80^\circ) + \sin^2(50^\circ) \end{aligned}$$

Exemple 2 :

$$\begin{aligned} A &= \cos^2(10^\circ) + \sin^2(40^\circ) + \cos^2(80^\circ) + \sin^2(50^\circ) \\ &= \sin^2(80^\circ) + \cos^2(50^\circ) + \cos^2(80^\circ) + \sin^2(50^\circ) \\ &= \sin^2(80^\circ) + \cos^2(80^\circ) + \cos^2(50^\circ) + \sin^2(50^\circ) \end{aligned}$$

Exemple 2 :

$$\begin{aligned}A &= \cos^2(10^\circ) + \sin^2(40^\circ) + \cos^2(80^\circ) + \sin^2(50^\circ) \\&= \sin^2(80^\circ) + \cos^2(50^\circ) + \cos^2(80^\circ) + \sin^2(50^\circ) \\&= \sin^2(80^\circ) + \cos^2(80^\circ) + \cos^2(50^\circ) + \sin^2(50^\circ) \\&= 1 + 1\end{aligned}$$

Exemple 2 :

$$\begin{aligned} A &= \cos^2(10^\circ) + \sin^2(40^\circ) + \cos^2(80^\circ) + \sin^2(50^\circ) \\ &= \sin^2(80^\circ) + \cos^2(50^\circ) + \cos^2(80^\circ) + \sin^2(50^\circ) \\ &= \sin^2(80^\circ) + \cos^2(80^\circ) + \cos^2(50^\circ) + \sin^2(50^\circ) \\ &= 1 + 1 \\ &= 2 \end{aligned}$$

Exemple 3 :

Exemple 3 :

$$B = \sin(80^\circ) + 7 \sin^2(50^\circ) - \cos(10^\circ) + 7 \sin^2(40^\circ)$$

Exemple 3 :

$$\begin{aligned} B &= \sin(80^\circ) + 7 \sin^2(50^\circ) - \cos(10^\circ) + 7 \sin^2(40^\circ) \\ &= \cos(10^\circ) + 7 \cos^2(40^\circ) - \cos(10^\circ) + 7 \sin^2(40^\circ) \end{aligned}$$

Exemple 3 :

$$\begin{aligned} B &= \sin(80^\circ) + 7 \sin^2(50^\circ) - \cos(10^\circ) + 7 \sin^2(40^\circ) \\ &= \cos(10^\circ) + 7 \cos^2(40^\circ) - \cos(10^\circ) + 7 \sin^2(40^\circ) \\ &= \cos(10^\circ) - \cos(10^\circ) + 7 \cos^2(40^\circ) + 7 \sin^2(40^\circ) \end{aligned}$$

Exemple 3 :

$$\begin{aligned} B &= \sin(80^\circ) + 7 \sin^2(50^\circ) - \cos(10^\circ) + 7 \sin^2(40^\circ) \\ &= \cos(10^\circ) + 7 \cos^2(40^\circ) - \cos(10^\circ) + 7 \sin^2(40^\circ) \\ &= \cos(10^\circ) - \cos(10^\circ) + 7 \cos^2(40^\circ) + 7 \sin^2(40^\circ) \\ &= 7 \left(\cos^2(40^\circ) + \sin^2(40^\circ) \right) \end{aligned}$$

Exemple 3 :

$$\begin{aligned} B &= \sin(80^\circ) + 7 \sin^2(50^\circ) - \cos(10^\circ) + 7 \sin^2(40^\circ) \\ &= \cos(10^\circ) + 7 \cos^2(40^\circ) - \cos(10^\circ) + 7 \sin^2(40^\circ) \\ &= \cos(10^\circ) - \cos(10^\circ) + 7 \cos^2(40^\circ) + 7 \sin^2(40^\circ) \\ &= 7 \left(\cos^2(40^\circ) + \sin^2(40^\circ) \right) \\ &= 7 \times 1 \end{aligned}$$

Exemple 3 :

$$\begin{aligned} B &= \sin(80^\circ) + 7 \sin^2(50^\circ) - \cos(10^\circ) + 7 \sin^2(40^\circ) \\ &= \cos(10^\circ) + 7 \cos^2(40^\circ) - \cos(10^\circ) + 7 \sin^2(40^\circ) \\ &= \cos(10^\circ) - \cos(10^\circ) + 7 \cos^2(40^\circ) + 7 \sin^2(40^\circ) \\ &= 7 \left(\cos^2(40^\circ) + \sin^2(40^\circ) \right) \\ &= 7 \times 1 \\ &= 7 \end{aligned}$$

Exemple 4 :

Exemple 4 :

$$C = \cos^2(15^\circ) + \cos^2(75^\circ) - 2 \tan(35^\circ) \times \tan(55^\circ)$$

Exemple 4 :

$$\begin{aligned}C &= \cos^2(15^\circ) + \cos^2(75^\circ) - 2 \tan(35^\circ) \times \tan(55^\circ) \\&= \sin^2(75^\circ) + \cos^2(75^\circ) - 2 \times \frac{1}{\tan(55^\circ)} \times \tan(55^\circ)\end{aligned}$$

Exemple 4 :

$$\begin{aligned}C &= \cos^2(15^\circ) + \cos^2(75^\circ) - 2 \tan(35^\circ) \times \tan(55^\circ) \\&= \sin^2(75^\circ) + \cos^2(75^\circ) - 2 \times \frac{1}{\tan(55^\circ)} \times \tan(55^\circ) \\&= 1 - 2\end{aligned}$$

Exemple 4 :

$$\begin{aligned}C &= \cos^2(15^\circ) + \cos^2(75^\circ) - 2 \tan(35^\circ) \times \tan(55^\circ) \\&= \sin^2(75^\circ) + \cos^2(75^\circ) - 2 \times \frac{1}{\tan(55^\circ)} \times \tan(55^\circ) \\&= 1 - 2 \\&= -1\end{aligned}$$

Application :

Application :

Simplifier les expressions suivantes :

i) $A = \cos^2(35^\circ) + \sin^2(33^\circ) + \sin^2(35^\circ) + \cos^2(33^\circ)$

ii) $B = \cos^2(15^\circ) + \cos^2(75^\circ) - 2 \tan(35^\circ) \times \tan(55^\circ)$

iii) $C = \sin(25^\circ) - \sin(65^\circ) + \cos(25^\circ) - \cos(65^\circ)$